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Research paper

Effects of winter climate parameters on the thermal performance of dynamic rotating latent-energy-storage envelope (DRLESE)

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A R T I C L E I N F O

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A B S T R A C T

The accurate application of the Dynamic Rotating Latent-Energy-Storage Envelope (DRLESE) system necessitates careful consideration of outdoor climate conditions, which significantly impacted latent heat operation of Phase-Change Materials. The effects of climate parameters were explored by considering five convective heat transfer coefficients ranging from 10 W/(m·K) to 30 W/(m·K), six daily solar radiation intensities ranging from 15.0 MJ/Day to 30.0 MJ/Day, six air average temperatures ranging from −10.0°C to 15°C, and six fluctuation ranges of air temperature ranging from 5°C to 20°C. The thermal performance of the DRLESE system was evaluated by employing the liquid fraction of Phase-Change Materials (PCM), the thermal quantity of DRLESE, and the inner surface heat flow. The numerical results demonstrated that climate parameters have a profound effect on the thermal performance of the DRLESE system. Enhancing convective heat transfer coefficient or lowering outdoor air temperature can significantly attenuate thermal performance by promoting convective thermal dissipation, while increasing daily solar radiation intensity can provide ample solar radiation for absorption by the DRLESE system. With convective heat transfer coefficient increased from 10 W/(m^2·K) to 20 W/(m^2·K), the daily solar radiation intensity increased from 15MJ/Day to 30MJ/Day, outdoor air temperature increased from −10°C to 15°C, the fluctuation range increased from 5°C to 20°C, indoor effective thermal release was increased −55.38 %, 204.66 %, 241.00 % and 8.11 %, respectively. In addition, employing the transparent covers and selecting the appropriate PCM were recommended to enhance thermal performance of the DRLESE system.

1. Introduction

Energy and environment are two major challenges faced by mankind (Liu, et al., 2022). Building energy consumption accounts for more than 35 % of the social energy consumption in China (Yao et al., 2024), and becomes the largest terminal part, so building energy conservation has a profound significance on the energy crisis alleviation and the environment protection (Liu et al., 2024a). In China, building energy consumption accounts for more than 30 % of the social energy consumption, and becomes the largest terminal part, so building energy conservation has a profound significance on the energy crisis alleviation and the environment protection (Xiao et al., 2024). The passive design of building envelopes has been recognized as a crucial factor in reducing the escalating energy consumption associated with air-conditioning and heating within buildings (Meng, 2024a). Phase change materials (PCMs) are widely used in buildings due to their exceptional energy density at constant (Liu et al., 2024b). The latent-energy-storage envelope is usually built by employing Form-stable composite PCMs as a wall layer or mixing the microencapsulated PCMs with traditional building materials (Meng, 2024b). However, different from traditional building materials, PCM can only exert its potential improvement on the thermal performance of buildings through latent heat storage and release when a phase change occurs (Gao and Meng, 2023). Consequently, the thermal performance of latent-energy-storage envelope is highly reliant on outdoor climate conditions (Murathan and Manioğlu, 2024).

1.1. For traditional latent-energy-storage envelopes

For the climate adaptation of traditional Static Latent-Energy-Storage envelopes, Liu et al. (2023) investigated the adaptability of PCM parameters and configurations for reducing energy demands in lightweight buildings under varying climates. Their findings revealed that local climate significantly affected PCM parameters and configurations. The payback period for the same amount of PCM applied in Japan was shorter compared to China, with a payback period ranging from 18.2 to 37.1 years for 5 mm PCM installed on roofs in China, and

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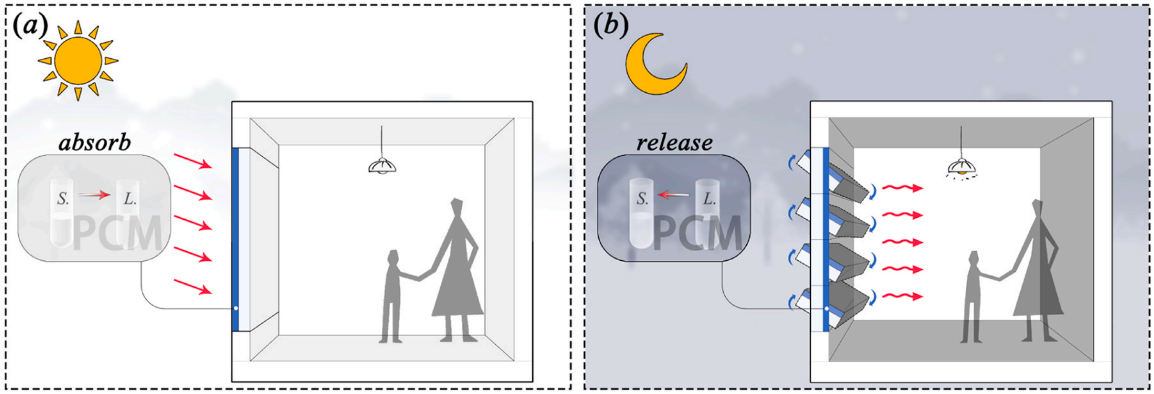

Fig. 1. Application diagram of the DRLESE system.

10.4–35.8 years for Japan. Adilkhanova et al. (2021) assessed the energy-saving potential of PCM based on diverse climatic conditions and demonstrated its effectiveness in reducing indoor discomfort; however, the extent of energy savings depended on specific climatic conditions. The research from Zwanzig et al. (2013) highlighted a strong correlation between weather patterns and PCM performance, emphasizing the importance of considering climatic regions when selecting appropriate thermal properties for PCMs. Salihi et al. (2022) placed the PCM layer in walls under the semi-arid climate and their numerical investigation showed that the suitable phase-change temperature was 27–29°C in a semi-arid climate and that compared to the single-layer PCM, employing the double or triple PCM layers was more efficient.

Abilkhassenova et al. (2023) performed environmental analysis for PCM-integrated buildings by considering all the fuel sources producing electricity in 60 cities, covering 15 climate zones. Their finding showed the integration of optimized PCM resulted in significant energy savings in each climate zone except equatorial climate, reducing maximum energy consumption up to 19,477 kWh (32.2 %) in the arid climate zone. Li et al. (2022) investigated the effects of walls containing PCM foamed cement on thermal performance and economy of single-story building in 5 different climate zones in China and found the energy saving rates are 20.17 % (Beijing), 15.87 % (Shenzhen), Shanghai (22.26 %) and 37.02 % (Kunming).

However, the conventional application of PCM is primarily focused on enhancing thermal inertia in buildings (Pirasaci and Sunol, 2024), thereby regulating heat transfer through envelopes and improving indoor thermal comfort (Refahi et al., 2024).

1.2. For the dynamic envelopes

To transition from passive thermal regulation to active heating, the dynamic envelopes have been proposed with the aim to address the limitations associated with traditional envelopes. Fawaier and Bokor (2022) demonstrated that dynamic thermal insulation can achieve over 40 % more savings compared to building envelopes with equivalent static thermal insulation. Zhou and Razaqpur (2022) developed a dynamic Trombe wall incorporating phase change material (PCM) and found that the use of dynamic rotation reduced thermal loss by 29 % during the discharge process, resulting in an overall increase in thermal efficiency by 20 %, compared to the static configuration. In another study, Zhou and Razaqpur (2024) analyzed the thermal performance of dynamic Trombe walls with and without PCM. They found that energy-saving efficiencies were higher by 25.3 % and 17.5 %, respectively, while their respective thermal efficiencies could reach up to 79.8 % and 35.4 %, when compared to their static counterparts.

The above performances show that the dynamic envelope has more energy-saving advantages than the traditional latent energy storage envelopes, but how to utilize the energy-saving potential of the system still needs to be studied. Through repositioning the PCM layer in relation to the insulation layer, Gracia (2019a) successfully achieves cooling storage during nighttime and cooling release during peak load hours. The study results showed that the phase change material melting temperature and turnover time significantly affected the energy-saving benefits, with the dynamic phase change material system reducing the cooling load by 379 % in the optimal configuration. Additionally, Gracia (2019b) investigates the indoor heating application of this dynamic system, and numerical results indicate a heating load reduction exceeding 100 % across all studied envelopes. The system functions as a thermal barrier, actively providing heating through stored solar energy. In addition, it was shown that the amount of PCM, melting temperature, and the weather condition impacts the energy-saving benefits significantly.

To effectively manage the dynamic adjustment of the PCM layer, Gracia et al. (2020) employed two intelligent control algorithms which maximize benefits and prevent potential malfunctions. Furthermore, Zhang et al. (2022) propose a dynamic PCM wall capable of storing passive solar energy during daytime and releasing it indoors at night. Their numerical results show that dynamic walls in Tianjin and Harbin save 89 % and 49 % of energy consumption, respectively, and indicate that different thicknesses of phase change material layers should be used for different climates.

In the study of the effect of climatic parameters on the thermal performance of the DRLESE system, Kishore et al. (2021) analyzed the effect of using dynamically flipped phase change walls in different cities, and the results showed that high temperatures in the summer and low temperatures in the winter significantly reduced the effectiveness of the flipped walls. Li et al. (2020) analyzed the conditions of applying the dynamic phase change roofing, and the simulation results showed that the system is suitable for buildings in cold and frigid climatic zones. In the study on the regulation strategy of the dynamic phase-change envelope, Gracia et al. (2020) found that low outdoor temperatures can make the system ineffective. However, none of the above studies have elucidated the specific effects of outdoor meteorological parameters on thermal performance of the dynamic envelope. Moreover, PCM can only exert its potential improvement on the thermal performance of buildings through latent heat storage and release when a phase change occurs (Meng, 2024b), so under this condition, outdoor climate conditions play a pivotal role in shaping the indoor built environment and significantly impact the practical effectiveness of latent-energy-storage envelope in buildings.

1.3. The purpose of this study

In this study, the Dynamic Rotating Latent-Energy-Storage Envelope (DRLESE) system was proposed in Fig. 1. As shown in Fig. 1, the PCM layer of the DRLESE system is in outdoors to enable the solar heat efficiently captured during the daytime. And then the PCM layer is positioned into indoor side by rotating Latent-Energy-Storage envelope, and

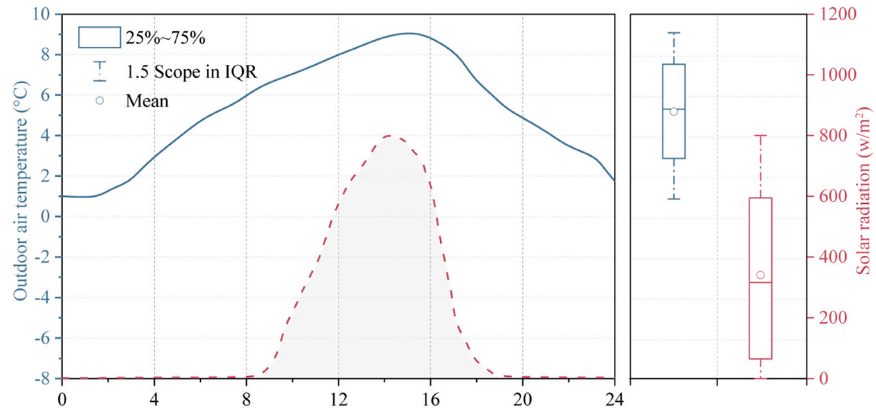

Fig. 2. Variation of outdoor air temperature and solar radiation.

the absorbed heat energy can be released directly into indoors to supply the indoor heating demand effectively. By the dynamic rotation, the DRLESE system can achieve the highly efficient absorption and release of solar energy, compared to traditional Latent-Energy-Storage envelopes.

In the thermal storage process in Fig. 1(*a*), the phase-change state of PCM and corresponding heat storage greatly relies on local climate environment, and then it affects indoor heat release highly in Fig. 1(*b*). Therefore, it is imperative to consider the local climate environment when applying the DRLESE system. As mentioned earlier, there have been numerous published studies on the climate adaptation of traditional Latent-Energy-Storage envelopes, but the DRLESE system is still unexplored. Compared to static latent-energy-storage envelopes, the DRLESE system can release the absorbed heat storage directly for indoor heating, so it is necessary to analyze the climate adaptation of the DRLESE system. Consequently, this study aims to analyze the effect of outdoor climate parameters including convective heat transfer coefficient, daily solar radiation intensity, average outdoor air temperature value and fluctuation range. The evaluation indexes employed are liquid fraction, total thermal storage and average heat flow. Numerical simulation is utilized and verified through the previous experiment to provide fundamental support for selecting suitable DRLESE systems at a local level. The DRLESE system can be applied to improve indoor thermal environment for the buildings without the heating or the heating energy consumption of conventional buildings in winter.

2. Description of investigation cases and the dynamic rotating latent-energy-storage envelope

In this section, the investigation cases are firstly described on winter climate parameters. Then, the typical DRLESE system is described with the clear control principles of rotation timings

2.1. Description of investigation cases

In this study, the thermal transfer is mainly considered and under this condition, the humidity transfer between wall and outdoor environment is ignored. The thermal transfer between wall and outdoor environment mainly relied on the convection heat transfer and solar heat gain, which involved the solar radiation, outdoor air temperature and wind speed in outdoor climate. It is noted that the wind speed had the high effect on convective heat transfer coefficient (CHTC) on the outer surface of buildings (Meng, 2024a). Therefore, in this study, the outdoor climate parameters primarily focus on four key parameters including CHTC on the outer surface, the daily solar radiation intensity (DSRI), and the average value and fluctuation range of outdoor air temperature (OAT-AV and OAT-FR). Fig. 2 gives the variation of outdoor air temperature and solar radiation in the typical day during winter (Nanjing city, China)

Table 1
Four investigation cases on analyzing the impact rules from outdoor climate parameters.

Climate parameters	Convective heat transfer coefficient (W/(m·K))	Daily solar radiation intensity (MJ/Day)	Outdoor air temperature (°C)	
			Average value	Fluctuation range
Case I	10, 15, 20, 25, 30	25.0	6.19	8.54
Case II	20	15.0, 17.5, 20.0, 25,0, 27.5, 30.0	6.19	8.54
Case III	20	25.0	−10, −5, 0, 5, 10, 15	8.54
Case IV	20	25.0	6.19	5.0, 7.5, 10.0, 12.5, 15.0, 17.5, 20

Under these climate conditions, OAT-AV and OAT-VR are recorded as 6.19°C and 8.54°C respectively, while the peak solar radiation intensity reaches approximately 800 W/m^2 with a daily solar radiation of 25.0 MJ/Day. Table 1 presents four investigation cases analyzing the impact rules from outdoor climate parameters for the DRLESE system.

- Case I considers the impacts of convective heat transfer coefficient ranging from 10 W/(m·K) to 30 W/(m·K), which is highly affected by indoor relative humidity, temperature difference between outer surface and outdoor air, outdoor wind speed, among other parameters.
- Case II examines the effects of daily solar radiation intensity varying from 15.0 MJ/Day to 30 MJ/Day, corresponding to peak solar radiation intensities ranging from 480 W/m^2 to 960 W/m^2.
- Cases III and IV investigate the effects from outdoor air temperature, which is determined by geographical latitude and local landforms.

The above cases encompass the prevailing winter climate conditions in major cities across China.

2.2. Description of the envelope model

The section diagram of the DRLESE in this study is presented in Fig. 3, which consists of a 1 mm metal sheet layer, a 50 mm PCM layer, a 100 mm Thermal Insulation Material (TIM) layer, and another 1 mm metal sheet layer. Additionally, Table 2 provides the thermal properties of the wall materials investigated in this study. Paraffin 27 was selected as the optimized phase-change material due to its cost-effectiveness, high latent heat capacity (178.5 J/g), and excellent performance

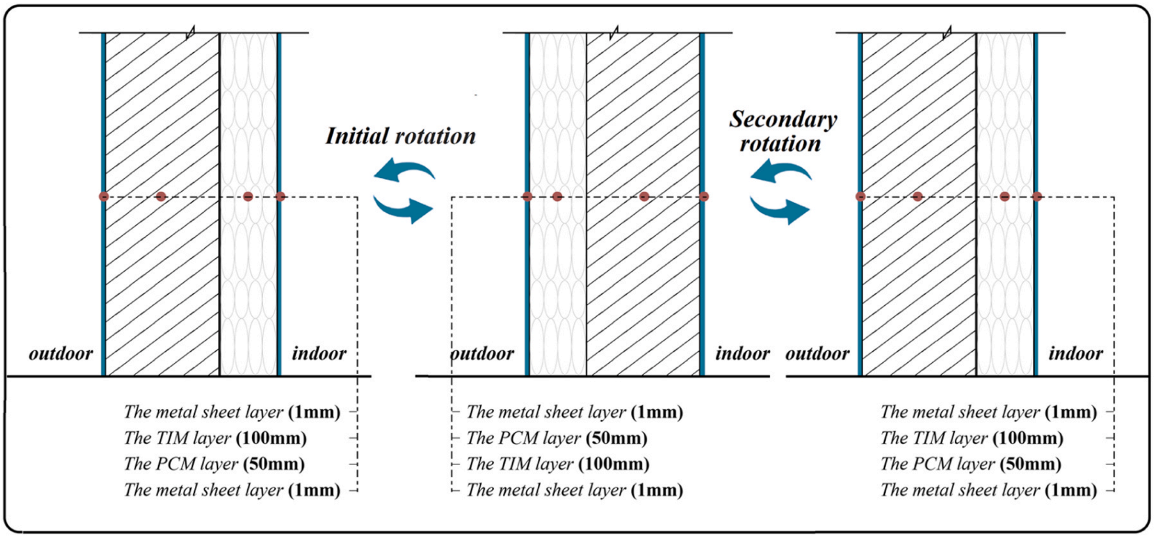

Fig. 3. Envelope diagram of the DRLESE system.

Table 2
Thermal properties of wall materials in this study.

Wall material	Specific heat (J/(kg•K))	Density (kg/m^3)	Thermal conductivity (W/(m•K))
Expanded polystyrene (EPS)	1330	20	0.037
Metal sheet	470	7966	13.31
PCM	1785	1300	2.0
Plaster	1050	1406	0.412

stability within the temperature range of 26 °C to 29 °C. Furthermore, incorporation of metal nanoparticles was employed to enhance the heat conductivity coefficient to 2.0 W/(m•K).

Additionally, according to the previous study (Meng, 2024c), the suitable rotation timings can adhere to the following control principles:

- The optimal initial rotation occurs when the inner surface temperature aligns with the sol-air temperature in the forenoon.
- The optimal secondary rotation is attained when the sol-air temperature equals that of PCM liquid in the afternoon.

3. Numerical model and its verification

3.1. Description of thermal transfer model

The presence of continuous wall heat transfer between the inner and outer surfaces indicates a two-dimensional heat transfer phenomenon. Moreover, the outer surface accounts for solar radiation gain and convection heat transfer, while only convection heat transfer occurs on the inner surface. Therefore, the heat transfer model can be mathematically described as follows:

$$\frac{\partial(\rho H)}{\partial t} = \frac{\partial}{\partial x}\left(\lambda\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda\frac{\partial T}{\partial y}\right) \tag{1}$$

Where, ρ, H and λ are the density, the enthalpy value and heat conductivity coefficient of wall materials, respectively.

For the traditional building material:

$$H = C_P T \tag{2}$$

For PCM

$$H_{PCM} = \begin{cases} \int_{T_0}^{T} C_{P,PCM} dT & (T < T_S) \\ \int_{T_0}^{T_S} C_{P,PCM} dT + \frac{T - T_S}{T_L - T_S} L_{PCM} & (T_S \le T \le T_L) \\ \int_{T_0}^{T_S} C_{P,PCM} dT + L + \int_{T_L}^{T} C_{P,PCM} dT & (T_L < T) \end{cases} \tag{3}$$

Where, ρ_{PCM}, $C_{P,PCM}$, $L_{PCM}$and λ_{PCM} are the density, the specific heat capacity, phase-change latent heat and heat conductivity coefficient of PCM, respectively. T_S and T_L are solidus and liquidus temperatures.

The thermal boundaries in outer and inner surface are described, respectively, as following:

$$-\lambda\frac{\partial T}{\partial x}\bigg|_{x=0} = h_{out}\left(T_{out} - T_{1,out}\right) + \alpha I \tag{4}$$

$$-\lambda\frac{\partial T}{\partial x}\bigg|_{x=\delta} = h_{in}\left(T_{1,in} - T_{in}\right) \tag{5}$$

Where, h_{in} and h_{out} are the convective heat transfer coefficients in inner and outer surfaces. T_{in} and T_{out} are indoor and outdoor air temperatures, while $T_{1,in}$ and $T_{1,out}$ are inner and outer surface temperatures. I denote solar radiation intensity and α is the radiation absorptivity.

The adiabatic boundaries were employed at top and below surfaces:

$$-\lambda\frac{\partial T}{\partial x}\bigg|_{y=0 \text{ or } H} = 0 \tag{6}$$

3.2. Numerical approach and independence tests

The above-mentioned equations are solved by commercial software of ANSYS FLUENT 16.1. The Finite Volume Method is utilized to discretize the above governing equations, and the computational region is divided into the uniform rectangular grids along the thickness and height direction of envelops. And the unsteady terms and the diffusion terms are interpolated by using the second-order upwind scheme. The convergence of the computations is declared at each time instant, when the following criterion is satisfied:

$$\frac{\sum T_{i,j}^{m+1} - T_{i,j}^{m}}{\sum \left|T_{i,j}^{n+1}\right|} \le 10^{-5} \tag{7}$$

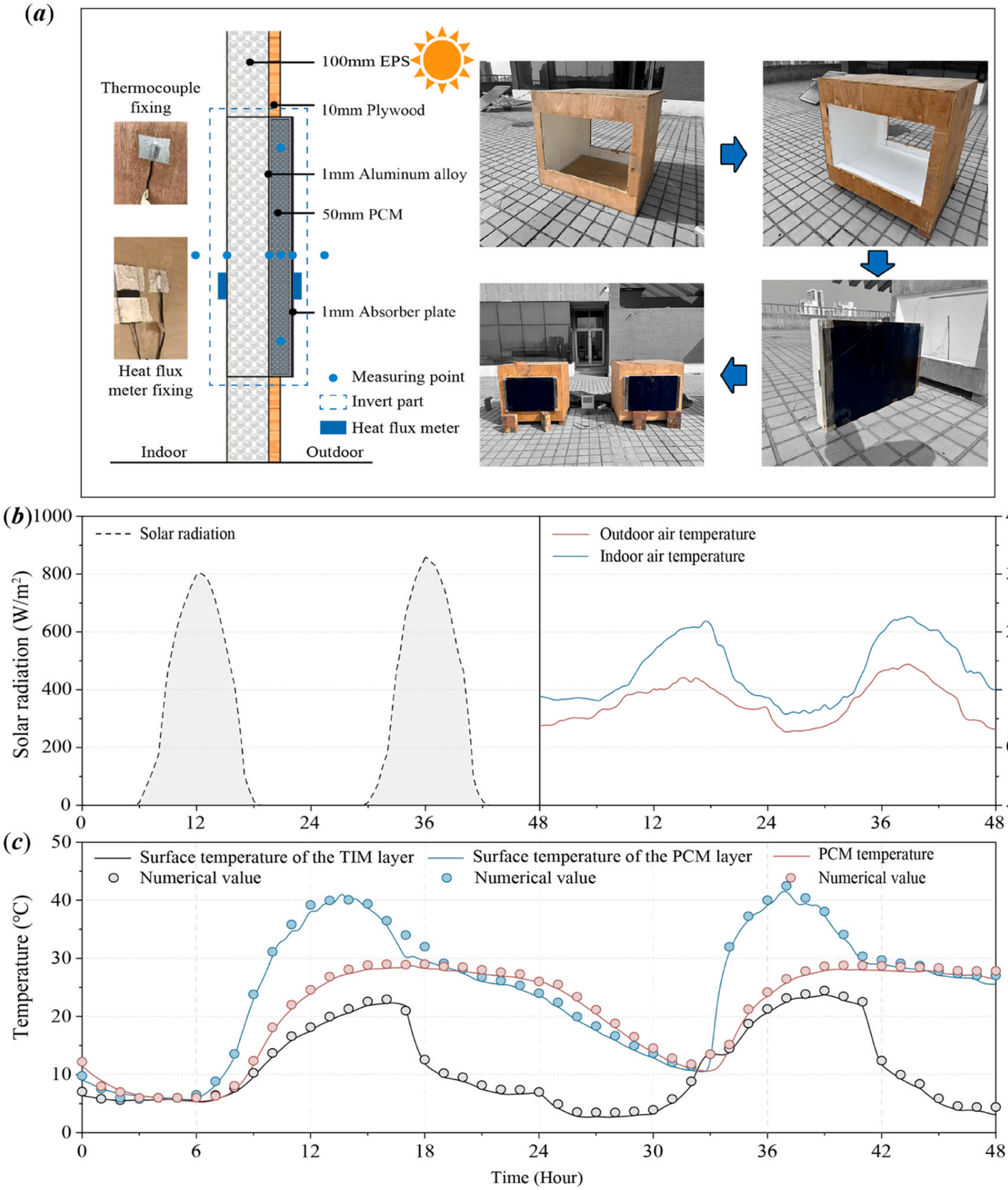

Fig. 4. (*a*) Experimental Model & measurement locations, (b) solar radiation, outdoor and indoor air temperatures, (c) comparison of numerical and experimental values.

where, *m* is the internal iteration number.

To select the optimal grid density and the reasonable time step, the sensitivity analysis was done with the grid densities of 10^{-5} m, 10^{-4} m and 10^{-3} m and the time steps of 0.1 s, 1 s and 10 s. By taking the grid density of 10^{-6} m and the time step of 0.01 s as the reference, the suitable values were considered as 10^{-3} m and 1 s for the grid density and the time step, respectively, with a relative error of 0.5 % as the determination criterion

3.3. *Description of the model verification*

To verify the numerical model, a building model was built in Fig. 4 (*a*) with the size of 1200 mm (Length)×1000 mm (width)×1000 mm (Hight) with an DRLESE system including from the 1 mm absorber plate, the 50 mm PCM layer plate, and the 100 mm TIM layer from outside into inside in the thermal absorbing process.

Table 3
Measuring range and accuracy of the instrument.

Instruments	Model	Measuring range	Accuracy
T-type thermocouples	TP1000	−20℃–70℃	±0.5℃
Solar radiometers	JTR05Z	0–2000 W/m^2	±2 %
Heat flux meter	JTRL−2	−500–500 W/m2	±3 %

Fig. 4(b) shows the variation of solar radiation, outdoor and indoor air temperatures during the experimental period. As shown in Fig. 4(b), it can be clearly seen that due to the DRLESE system, indoor air had the high temperature in the building model, compared to outdoor air. The measuring equipment used for the experiment is shown in Table 3. All instruments had been verified to ensure experimental accuracy before

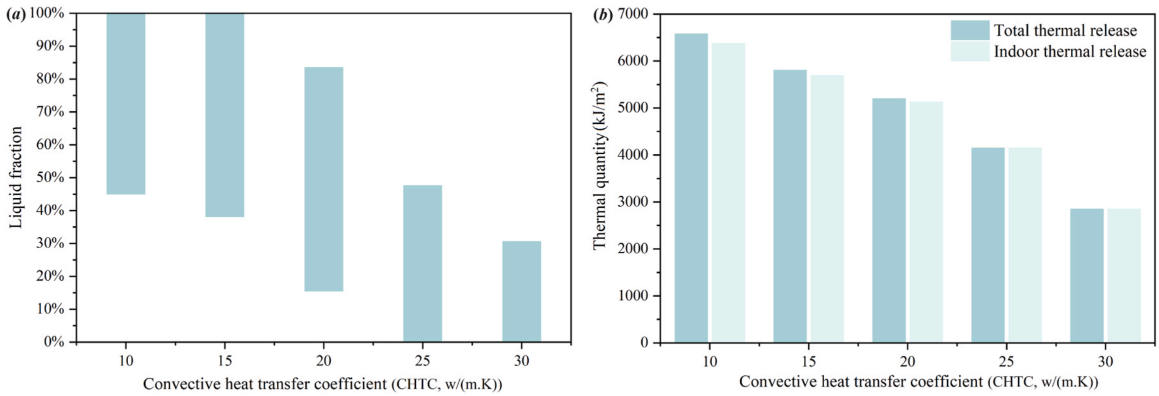

Fig. 5. Variation of (*a*) liquid fraction and (*b*) thermal quantity with convective heat transfer coefficient.

the experiment, and they recorded data at a frequency of every 10 minutes.

The experiment was conducted from March 28 to March 29 in Qingdao city (36°10′ latitude and 120°37′ longitude), China. The solar radiometer is installed on the south wall and parallel to the DRLESE and records the intensity of incident solar radiation.

In the validation process, the uncertainty for the experimental facility resulted from the measurement error. It could be calculated by the uncertainty analysis method as the following equation (Ge et al., 2024):

$$\varepsilon = \sqrt{\left(\frac{\partial f}{\partial x_1}U_1\right)^2 + \left(\frac{\partial f}{\partial x_2}U_2\right)^2 + \cdots + \left(\frac{\partial f}{\partial x_n}U_n\right)^2} \tag{8}$$

The uncertainty of the thermocouple is 1.25 %, the uncertainty of the solar radiometer is 4.13 %, and the uncertainty of the heat flow meter is 3.21 %. The overall uncertainty of the experiment is 5.38 %.

The thermal boundaries in Fig. 4(b) were employed for the thermal transfer model, and Fig. 4(c) compared numerical and experimental values. As shown in Fig. 4(c), there exists a consistent trend between the present values and published ones, with a maximum relative error lower than 2.91 % for inner surface temperatures, indicating high accuracy of the above heat transfer model.

3.4. Description of the evaluation parameters

The DRLESE system is evaluated from three aspects. The liquid fraction (L) reflects the melting and solidifying states of PCM in the DRLESE system and its numerical equation is following:

$$L = \begin{matrix} 100\% & (T > T_L) \\ (T - T_S)/(T_L - T_S) & (T_S \le T \le T_L) \\ 100\% & (T < T_S) \end{matrix} \tag{9}$$

The total thermal storage (Q_{TTS}) represents the solar gains obtained from PCM with its numerical equation as following:

$$Q_{TTS} = \int_0^{86400} -\lambda \frac{\partial T}{\partial x}\bigg|_{x=152mm} dt \left(-\lambda \frac{\partial T}{\partial x}\bigg|_{x=152mm} > 0\right) \tag{10}$$

The total thermal release (Q_{TTR}) represents cumulative thermal emissions of Q_{TTS}, a portion of which is released indoors for active indoor heating and defined as indoor effective thermal release (Q_{IETR}). Their numerical definitions are following:

$$Q_{TTR} = \int_0^{86400} -\lambda \frac{\partial T}{\partial x}\bigg|_{x=152mm} dt \left(-\lambda \frac{\partial T}{\partial x}\bigg|_{x=152mm} < 0\right) \tag{11}$$

$$Q_{IETR} = \int_{t_l}^{t_S} -\lambda \frac{\partial T}{\partial x}\bigg|_{x=152mm} dt \tag{12}$$

According to Q_{TTR} and Q_{IETR}, the thermal efficiency (η) is following:

$$\eta = \frac{Q_{IETR}}{Q_{TR}} = \frac{\int_{t_l}^{t_S} -\lambda \frac{\partial T}{\partial x}\Big|_{x=152mm} dt}{\int_0^{86400} -\lambda \frac{\partial T}{\partial x}\Big|_{x=152mm} dt \left(-\lambda \frac{\partial T}{\partial x}\Big|_{x=152mm} < 0\right)} \times 100\% \tag{13}$$

The inner surface heat flow displays the heat flow from inner surface to indoor air and its transient and average values (q and q_{AVG}) can be defined as following:

$$q = h_{in}\left(T_{in} - T_{1,in}\right) = \begin{cases} -\lambda \frac{\partial T}{\partial x}\Big|_{x=152mm} & (0 \le t \le t_I) \\ -\lambda \frac{\partial T}{\partial x}\Big|_{x=0mm} & (t_I < t < t_S) \\ -\lambda \frac{\partial T}{\partial x}\Big|_{x=152mm} & (t_S \le t \le 86400) \end{cases} \tag{14}$$

$$q_{AVG} = \frac{\int_0^{86400} q dt}{86400} = \frac{\int_0^{86400} h_{in}\left(T_{in} - T_{1,in}\right) dt}{86400} \tag{15}$$

4. Results and discussion

This study considers four climate parameters (CHTC, DSRI, OAT-AV and OAT-VR), while utilizing liquid fraction, total thermal storage and average heat flow as evaluation indices of the DRLESE.

4.1. Effect of convective heat transfer coefficient on thermal performance of DRLESE

The variation of the liquid fraction with convective heat transfer coefficient is shown in Fig. 5(a). It can be observed that both the peak and lowest liquid fractions gradually decrease as the convective heat transfer coefficient increases. PCM can fully melt with convective heat transfer coefficients of 10 W/(m·K) and 15 W/(m·K), while it can completely solidify with convective heat transfer coefficients of 25 W/(m·K) and 30 W/(m·K). The convective heat transfer coefficient represents heat-transfer intensity between the outer surface and outdoor air. Due to the low temperature of outdoor air, the lower convective heat transfer coefficient reduces the thermal dissipation near the outdoors, favoring the higher heat storage to melt PCM. Conversely, the higher convective heat transfer coefficient promotes thermal dissipation between the PCM layer and outdoor air.

From the thermal quantity from Fig. 5(*b*), it is evident that the total thermal storage and indoor effective thermal release significantly decrease with an increase in the convective heat transfer coefficient, indicating that a higher convective heat transfer coefficient does not enhance the thermal performance of the DRLESE system. This phenomenon can be attributed to the improved convectional thermal

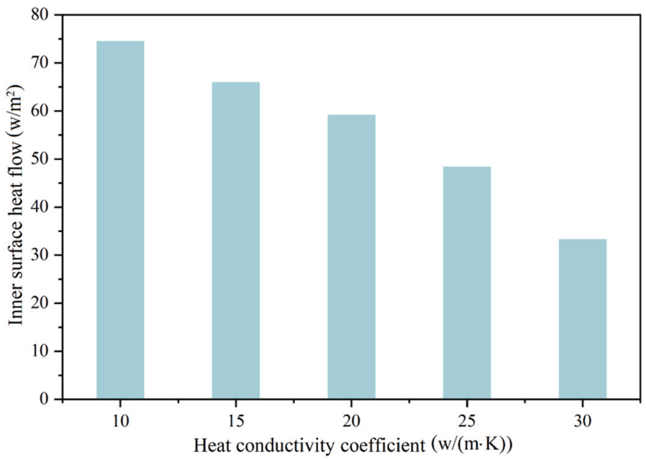

Fig. 6. Variation of inner surface heat flow with convective heat transfer coefficient.

Table 4
Comparison of thermal quantity and heat flow under the different convective heat transfer coefficients.

Evaluation parameter	Convective heat transfer coefficient (w/(m·K))				
	10	15	20	25	30
Total thermal release (kJ/m^2)	6572.81	5801.54	5192.53	4144.06	2844.77
Indoor thermal release (kJ/m^2)	6369.66	5687.71	5123.69	4143.06	2843.97
Inner surface heat flow (w/m^2)	74.38	65.83	59.06	48.26	33.19

dissipation due to the presence of the PCM layer near outdoors, resulting in the reduction of total thermal storage and subsequently lowering indoor effective thermal release. In all cases, the thermal efficiencies exceed 99 %, which showed the stored heat has been used efficiently. Moreover, increasing the convective heat transfer coefficient from 10 W/(m·K) to 20 W/(m·K) results in reducing thermal quantity from 6369.66 kJ/m^2 to 5123.69 kJ/m^2, which showed a reduction of indoor effective thermal release by 24.32 %.

The variation of inner surface heat flow with convective heat transfer coefficient is presented in Fig. 6 and this heat transfer occurs from inner surface to indoor air, facilitating active heating within the indoor environment. It can be observed that the inner surface heat flow gradually decreases as the convective heat transfer coefficient increases. Moreover, an increase in the convective heat transfer coefficient from 10 W/(m·K) to 20 W/(m·K) results in lowering inner surface heat flow from 74.38 w/m^2 to 59.05 w/m^2, showing a reduction of 25.94 % in inner surface heat flow.

Table 4 compared the thermal quantity and inner surface heat flow under the different convective heat transfer coefficients. Therefore, lowering convective heat transfer coefficient is in favor of the thermal performance improvement of the DRLESE system, and thereby lowering the wind speed around the exterior wall or adding the glass cover is the suitable technology.

4.2. Effect of daily solar radiation intensity on thermal performance of DRLESE

The daily solar radiation intensity serves as the primary heat source for the DRLESE system, so the high solar radiation intensity can provide the larger thermal storage and the higher operation temperature, which maybe need the high phase-change temperature and the more quantity of PCM. To address this problem, Fig. 7(a) illustrates the variation of the liquid fraction in response to changes in daily solar radiation intensity. It is observed that both the peak and lowest liquid fractions gradually decrease with increasing daily solar radiation intensity. The high daily solar radiation intensity can provide the more heat energy for the latent heat storage of the PCM layer and subsequently the liquid fraction enhancement.

The total thermal storage and indoor effective thermal release in Fig. 7(*b*) clearly exhibit a significant increase with the rise in daily solar radiation intensity, despite the limited variation range of liquid fraction. Moreover, as the daily solar radiation intensity increases from 15MJ/Day to 30MJ/Day, the thermal quantity increased from 1940.52 kJ/m^2 to 5934 kJ/m^2, which signified there is a remarkable 218.45 % increase

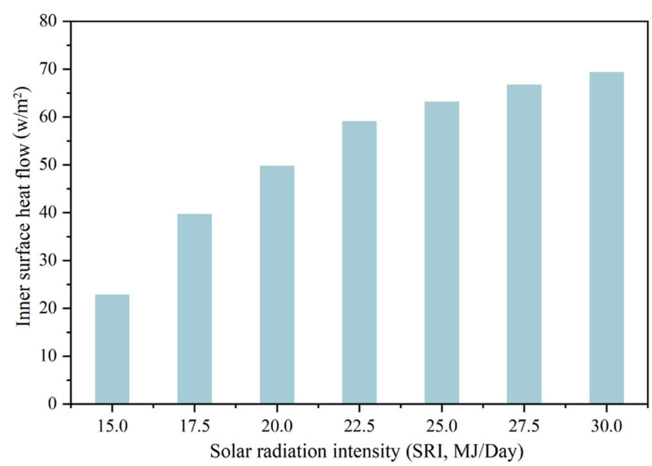

Fig. 8. Variation of inner surface heat flow with daily solar radiation intensity.

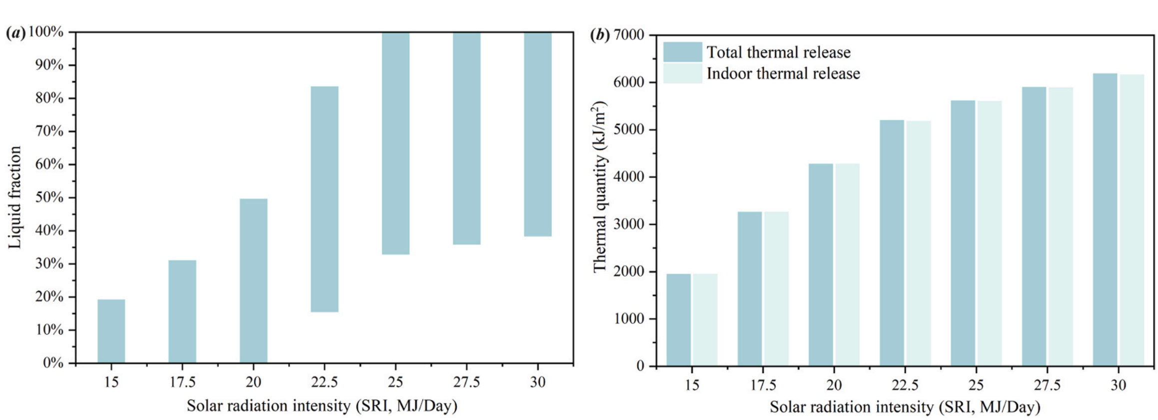

Fig. 7. Variation of (a) liquid fraction and (b) thermal quantity with daily solar radiation intensity.

Table 5
Comparison of thermal quantity and heat flow under the different daily solar radiation intensities.

Evaluation parameter	Daily solar radiation intensity (MJ/Day)						
	15.0	17.5	20.0	22.5	25	27.5	30.0
Total thermal release (kJ/m^2)	1940.52	3256.23	4267.67	5195.65	5609.70	5923.21	6179.73
Indoor thermal release (kJ/m^2)	1940.52	3196.23	4267.67	5103.23	5563.29	5823.65	5954.34
Inner surface heat flow (w/m^2)	22.72	40.12	49.71	59.35	63.08	67.23	69.23

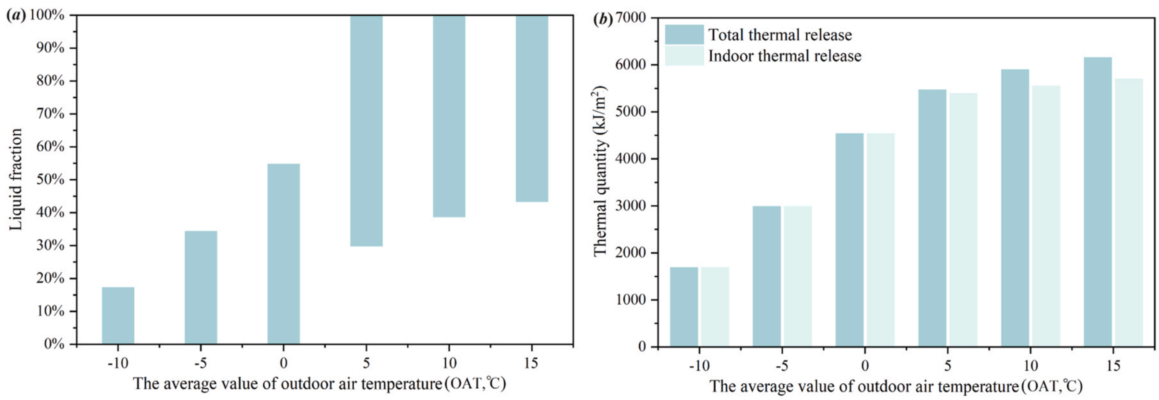

Fig. 9. Variation of (a) liquid fraction and (b) thermal quantity with outdoor average air temperature.

in indoor effective thermal release. Moreover, with increasing daily solar radiation intensity, the contribution efficiency is lowering from the intensity increase. It highlighted the high application efficiency of the DRLESE system under the higher solar radiation intensities. In all cases, the thermal efficiencies exceed 99 %, which showed the stored heat has been used efficiently.

The variation of inner surface heat flow with daily solar radiation intensity is presented in Fig. 8. It is evident that the inner surface heat flow increases significantly as the daily solar radiation intensity increases, despite the limited range of liquid fraction variation. Furthermore, an increase in daily solar radiation intensity from 15MJ/Day to 30MJ/Day results in increasing inner surface heat flow from 22.72 w/m^2 to 69.23 w/m^2, showing a 204.66 % increase in inner surface heat flow. Hence, the high solar radiation intensity promotes active heating of the DRLESE system. Table 5 compared the thermal quantity and inner surface heat flow under the different daily solar radiation intensities.

4.3. *Effect of outdoor average air temperature on thermal performance of DRLESE*

The outdoor air temperature serves as a crucial evaluation parameter for assessing the outdoor thermal environment. The low air temperature causes the thermal loss from DRLESE in the thermal charge process, so a transparent cover or solar house features is even recommended to mitigate the impact of thermal operation caused by the low temperature of outdoor air. To reveal the effect of outdoor air average temperature. Fig. 9(*a*) illustrates the relationship between the liquid fraction and outdoor average air temperature. It is evident that both the peak and minimum values of the liquid fraction gradually increase with rising the average temperatures. PCM fully solidifies with the average outdoor air temperature below 0 °C, while it completely melts when exceeding 5 °C. This

phenomenon can be attributed to the direct effect of outdoor average temperature on convective thermal dissipation. An elevation in outdoor average temperature reduces the temperature difference between outer surface and outdoor air, thereby diminishing convective thermal dissipation and consequently increasing the liquid fraction.

The total thermal storage and indoor effective thermal release in Fig. 9(*b*) exhibit a clear increase with rising outdoor average air temperature, particularly when the temperature is below 5 °C. Moreover, as the daily solar radiation intensity increases from −10 °C to 15 °C, the thermal quantity increased from 1682.20 kJ/m^2 to 5694.10 kJ/m^2, which showed there is a remarkable 265.53 % enhancement in indoor effective thermal release. It highlighted the high application efficiency of the DRLESE system in regions characterized by relatively higher winter temperatures.

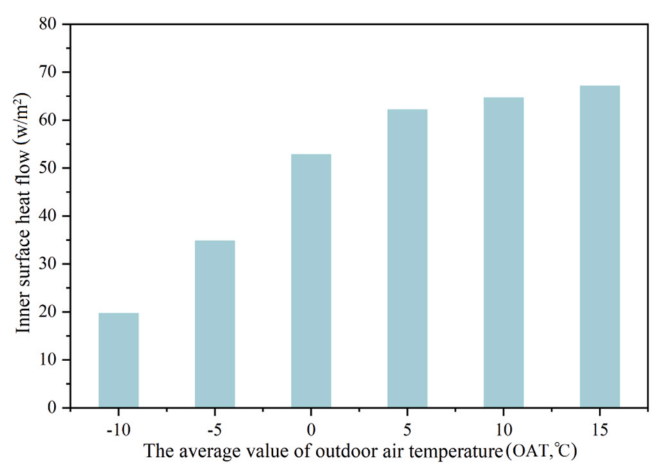

Fig. 10. Variation of inner surface heat flow with outdoor average air temperature.

The variation of inner surface heat flow with outdoor average air temperature is presented in Fig. 10. It is evident that the heat flow increases significantly as the outdoor average air temperature rises. Moreover, an increase in outdoor average air temperature from −10 °C to 15 °C results in increasing inner surface heat flow from 19.65w/m^2 to 67.02 w/m^2, signifying a 241.00 % increase in inner surface heat flow and under the lower outdoor average air temperature, there is the higher increase efficiency. Therefore, the application efficiency of the DRLESE system exhibits a strong dependence on outdoor average air temperature. Table 6 compared the thermal quantity and inner surface heat flow under the different outdoor average air temperatures.

Table 6
Comparison of thermal quantity and heat flow under the different outdoor average air temperatures.

Evaluation parameter	Outdoor average air temperature (°C)					
	-10	-5	0	5	10	15
Total thermal release (kJ/m^2)	1682.19	2980.91	4529.34	5463.62	5891.89	6148.9
Indoor thermal release (kJ/m^2)	1682.19	2980.91	4529.34	5384	5541.58	5694.1
Inner surface heat flow (w/m^2)	19.65	34.71	52.71	62.10	64.58	67.02

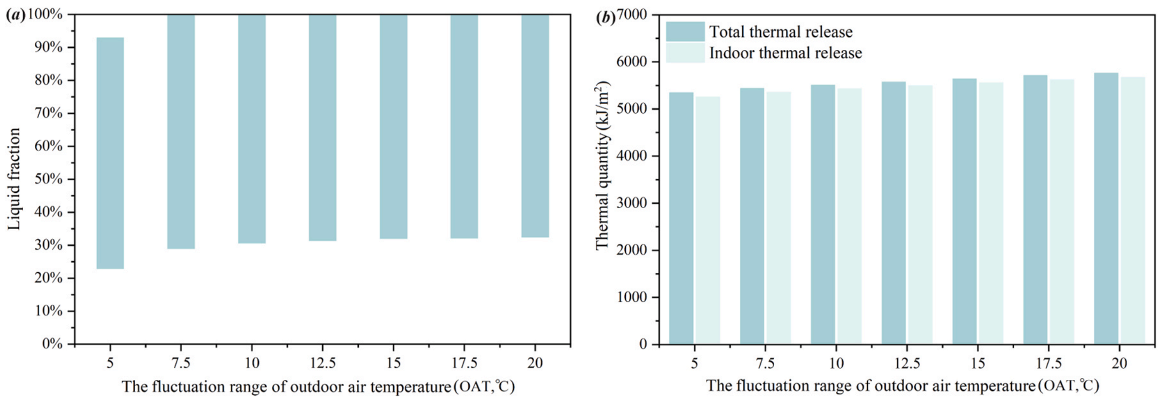

Fig. 11. Variation of (a) liquid fraction and (b) thermal quantity with the variation range of outdoor air temperature.

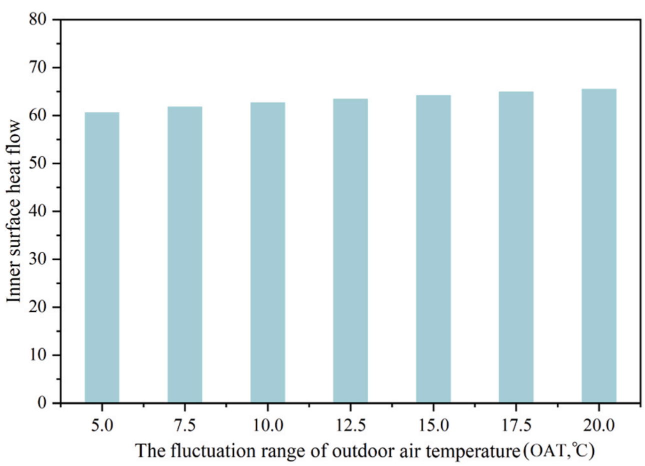

Fig. 12. Variation of inner surface heat flow with the variation range of outdoor air temperature.

4.4. Effect of temperature fluctuation range of outdoor air thermal performance of DRLESE

The outdoor air temperature undergoes diurnal fluctuations, with the highest temperatures at noon and the lowest temperatures at night due to solar radiation. Under the certain average temperature, the high temperature fluctuation can increase outdoor air temperature in the daytime, weakening the heat loss from the DRLESE, and lower outdoor air temperature in the nighttime, which has the low influence on the DRLESE system, due to the PCM layer rotated into indoors. To address the influence from temperature fluctuation range, the variation of the fluctuation range of outdoor air temperature is illustrated in Fig. 11(a). It can be observed that an increase in the variation range of outdoor air temperature leads to a higher liquid fraction. Specifically, as the variation range of outdoor air temperature increases, the daytime temperatures rise, and nighttime temperatures decrease. This phenomenon enhances solar radiation gain through the PCM layer near outdoors during the day, resulting in a peak liquid fraction reaching 100 %. The total thermal storage and indoor effective thermal release in Fig. 11(b) exhibit a gradual increase as the fluctuation range of outdoor air temperature increases. Moreover, it is observed that the indoor effective thermal release experiences a 6.01 % enhancement when the fluctuation range rises from 5 °C to 20 °C.

Fig. 12 illustrates the variation of inner surface heat flow in response to fluctuations in outdoor air temperature. It is evident that there is a significant increase in heat flow with greater fluctuations in outdoor air temperature. Specifically, as the fluctuation range increases from 5.0 °C to 20.0 °C, there is an observed increase of 8.11 % in inner surface heat flow. Table 7 compared the thermal quantity and inner surface heat flow under the different the variation ranges of outdoor air temperature.

5. Conclusions

The effect of outdoor climate on the thermal performance of the DRLESE system was analyzed, considering four parameters of convective heat transfer coefficient, daily solar radiation intensity, average value and fluctuation range of outdoor air temperature. A numerical thermal transfer model is constructed and experimentally validated, while the liquid fraction, total thermal storage, and average heat flow are employed as evaluation indices. The main conclusions are as follows:

Table 7
Comparison of thermal quantity and heat flow under the different variation ranges of outdoor air temperature.

Evaluation parameter	Variation range of outdoor air temperature (°C)						
	5.0	7.5	10.0	12.5	15.0	17.5	20.0
Total thermal release (kJ/m^2)	5340.23	5431.38	5504.47	5569.79	5635.90	5706.45	5758.03
Indoor thermal release (kJ/m^2)	5249.70	5349.30	5425.36	5488.55	5549.49	5617.18	5662.70
Inner surface heat flow (w/m^2)	60.48	61.69	62.60	63.36	64.10	64.84	65.39

- Increasing the exterior surface flow heat transfer coefficient reduces the amount of heat stored in the DRLESE system during the winter months, and when the convective heat transfer coefficient is increased from 10 W/(m^2·K) to 20 W/(m^2·K), the indoor heat release is reduced by 24.32 % and the interior surface heat flow is reduced by 25.94 %.
- The intensity of solar radiation significantly affects the amount of heat stored in the DRLESE system, with the effective indoor heat release and internal surface heat flow increasing by 218.45 % and 204.66 %, respectively, when solar radiation increases from 15 MJ/Day to 30 MJ/Day.
- An increase in the average outdoor temperature reduces outdoor heat dissipation and external surface heat flow by 265.53 % and 241.00 %, when the outdoor temperature increases from −10 °C to 15 °C.
- Increasing the range of outdoor temperature fluctuations can increase indoor heat release and internal surface heat flow. When the range of temperature fluctuations increased from 5 °C to 20 °C, the effective indoor heat release and heat flow increased by 6.01 % and 8.11 %, respectively.

Limitations and further work

The impact of winter climate parameters on the thermal performance of DRLESE has been analyzed in this study, yielding quantitative results, but it fails in the interact analysis of climate parameters and the more systematic evaluation need be done. Moreover, the potential improvement measures are recommended, but these technological interventions are based on the current numerical findings and require scientific verification and systematic evaluation in future studies.

CRediT authorship contribution statement

Xi Meng: Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Zihe Wang:** Methodology, Investigation, Data curation, Conceptualization. **Wenkai Fu:** Validation, Software, Methodology, Investigation. **Xudong Xie:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. And there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

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